

Education

- 2016 – 2018 **Oregon State University**, 🎓 *M.Sc.*, Computer Science [GPA: 4.0/4.0].
2010 – 2013 **Peking University, China**, 🎓 *M.Sc.*, Electronics Engineering [GPA: 3.71/4.0].
2006 – 2010 **Peking University, China**, 🎓 *B.Sc.*, Physics [GPA: 3.49/4.0].

Work Experience

Jul 2025 – **Senior Applied Scientist.**

Now **AMAZON AGI, SUNNYVALE, CA**

- Focus on Nova multimodal foundation models, full-scale reasoning RL, curriculum post-training, and customizations on Nova Forge.

A1 **Nova Omni & Nova-Lite: reasoning RL across dense and MoE models.**

- Executed full-scale production reasoning RL runs across 45B Dense and 33B-A4B MoE architectures, systematically scaling incremental context windows across 8K, 16K, and 32K stages with comprehensive datamix and method ablations.
- Incorporated SFT-replay and custom reward-function filters for 16K reasoning stage, effectively eliminating policy collapse and ensuring calibrated, high-quality generation across long reasoning trajectories.
- Delivered significant multimodal & reasoning gains on the final release model: achieved +38% on *Tau2*, +64% on *MMAgent*, +7.9% on *VMME*, +13% on *ChartUnderstanding*, and +4% on *STEM* benchmarks, while propelling Nova-Lite to top rankings on open arenas.

A2 **Curriculum reasoning RL strategy & unified post-training recipe optimization.**

- Unified multi-stage curriculum reasoning RL (8K → 32K) into a single-stage recipe via dynamic stage-mixing and Generalized BC, maintaining zero benchmark regression while standardizing cross-team protocols.
- Integrated On-Policy Distillation (OPD) to eliminate covariate shift via real-time teacher guidance, substantially accelerating convergence and stabilizing long-chain multi-step reasoning.
- Benchmarked speculative decoding under high concurrency; identified draft acceptance bottlenecks and established fail-fast compute guardrails, preventing hundreds of wasted GPU hours.

A3 **Amazon Nova Forge: multimodal SFT and RL customization for frontier models.**

- Led the technical design and execution of end-to-end multimodal training pipelines, standardizing modular recipes for enterprise-level frontier model customization across both SFT and RL post-training stages.
- Architected scalable post-training workflows supporting dynamic multimodal data packing, instruction alignment, and domain-adapted preference optimization (DPO/RLHF) tailored to diverse 3P enterprise requirements.
- Established a comprehensive automated evaluation harness and quality guardrail suite for customized models, ensuring substantial domain-specific capability gains while maintaining zero regression on general benchmarks.

Oct 2022 – **Senior Research Scientist.**

Jul 2025 **BYTEDANCE SEED, SAN JOSE, CA**

- Focus on Code LLM and multi-modal pretrained models

B1 **Seed-Coder base: pretraining dataset construction for base code LLM.**

- Built a data deduplication and decontamination pipeline to minimize human involvement in constructing code LLM pretraining data.
- Designed a high-quality code data filtering pipeline using a scoring model (distilled from a 33B model across 10 programming languages, generating 150k labeled samples for filter training), achieved outstanding *F1* and *RMSE*, enabling scalable filtering of 1.2TB curated code data for pretraining.

- Pretrained a 7B model on 2T tokens of high-quality code corpus, along with long-context training by supporting sequences of up to 32K tokens during the annealing phase, outperformed same-scaled *DeepSeek-Coder* model on *HumanEval* and *MBPP*, demonstrating superior data quality impact.
- B2 **Seed-Coder instruct and reasoning: post-training for code LLM.**
 - Developed supervised fine-tuning (SFT) frameworks, to support flexible deployment and to adapt our base models to various real-world tasks, followed by Direct Preference Optimization (DPO) to enhance specific capabilities such as code generation and reasoning.
 - Conducted diversity-focused data construction, difficulty & quality-focused data filtering, sandbox verification along with DPO training; achieved significant improvement over same-scaled *Coder-Instruct* models from *DeepSeek* and *Qwen2.5* in terms of *HumanEval*, *NaturalCodeBench*, *LiveCodeBench*, etc.
- B3 **TnS (Trust-and-Safety) Moderation: Multi-Modal Pretraining.**
 - Optimized a multi-platform data pipeline with efficient data packing and frame extraction, achieving 2× throughput over legacy setups.
 - Applied incremental multi-modal pretraining and tri-modal (video+text+audio) models, improving Violation *AUC* and *bF1@1*.
 - Evaluated generalization across dataset domains; enhancing *CLIP* model as SOTA on full TnS data.
- B4 **Tiktok Shop E-commerce AI: Multilingual & Multimodal Pretraining for Price Comparison.**
 - Scaled multilingual-multimodal dataset to 1B samples, training *CLIP* with *mBERT-Swin* and *XLNet* text tower, obtained significant gains on downstream tasks.
 - Boosted *recall* by +5% for multilingual retrieval at 100M data scale, and improved multimodal retrieval *P80* by +4.5%.
- Jun 2018 – **Research Scientist - Staff Research Scientist.**
- Oct 2022 **BAIDU RESEARCH, SUNNYVALE, CA**
 - 1 business product delivered, 2 applications deployed, 10 conference and journal papers on NLP and Bioinformatics published (including a featured paper accepted by **Nature**), and 3 patents licensed.
- C1 **STACL: Simultaneous Translation with Integrated Anticipation and Controllable Latency.**
 - A novel prefix-to-prefix framework for simultaneous translation that implicitly learns to anticipate in a single translation model, and an effective wait-k policy trained for target sentence concurrency.
 - ACL 2019 acceptance and patented. Featured on [worldwide media coverage](#), e.g., CNBC, FORTUNE.
- C2 **Incremental Text-to-Speech Synthesis with Prefix-to-Prefix Framework.**
 - The first neural incremental TTS approach based on prefix-to-prefix framework. Speech is synthesized in an online fashion, playing a segment of audio while generating the next, O(1) over O(n) latency, keeping naturalness as full sentence method.
 - Paper published at EMNLP 2020. Synthesized demo audios can be found on [inctts.github.io](#)
- C3 **LinearDesign: an efficient algorithms for Optimized mRNA Sequence Design.**
 - A surprisingly high efficient solution from computational linguistics to jointly optimize Messenger RNA (mRNA) vaccines' stability and codon usage, to tackle the critical issue of mRNA instability and degradation. [LinearDesign](#) takes only 11 min. for the COVID-19 Spike protein, substantially improve mRNA half-life and protein expression in vitro, and increase antibody response by up to 23× in vivo.
 - Lead commercialization with 3 contracts. 1 patent licensed and 1 paper published at **Nature**.
- C4 **CoV-Seq: a New Tool for SARS-CoV-2 Genome Analysis and Visualization.**
 - Developed CoV-Seq as automatic pipeline for fast and easy analysis of custom SARS-CoV-2 sequences. It predicts gene boundaries and identifies genetic variants, which are displayed in an interactive genome visualizer and are downloadable for further analysis. A weekly updated database of genetic variants of all publicly accessible SARS-CoV-2 sequences is also provided
- C5 **LinearFold: linear-time approximate RNA folding by 5'-to-3' dynamic programming and beam search.**
 - LinearFold is the first approximate algorithm in RNA folding to achieve linear runtime (and linear space) without imposing constraints on the output structure such as base-pair distance.
 - Merged status of intermediate statuses with beam size to compress the size of stacks.
- C6 **AutoSimTrans: 3-straight-year workshop on Automatic Simultaneous Translation.**
 - Lead Program Committee to host [Workshops & Challenges](#) at ACL2020, NAACL2021 and NAACL2022.
 - In charge of data preparation, submission pipeline and judgement system for Shared Task Challenge.
- July 2013 – **Engineer & Research program executive.**
- June 2016 **CHINA ELECTRIC POWER RESEARCH INSTITUTE (CEPRI), STATE GRID COOPERATION OF CHINA**
 - 2 conference papers published, and 3 patents authorized.

Selected Publications

- [1] Yuyu Zhang, ..., **Kaibo Liu**, et al., *Seed-Coder: Let the Code Model Curate Data for Itself*, arXiv preprint arXiv:2506.03524, 2025.
- [2] Jiaze Chen, ..., **Kaibo Liu**, et al., *Seed1.5-Thinking: Advancing Superb Reasoning Models with Reinforcement Learning*, arXiv preprint arXiv:2504.13914, 2025.
- [3] He Zhang, Liang Zhang, Ang Lin, Congc Xu, Ziyu Li, **Kaibo Liu**, David H. Mathews, and Liang Huang, *Algorithm for optimized mRNA design improves stability and immunogenicity [J]*, Nature, 2023.
- [4] He Zhang, Liang Zhang, Ziyu Li, **Kaibo Liu**, Boxiang Liu, David H. Mathews, and Liang Huang, *LinearDesign: Efficient Algorithms for Optimized mRNA Sequence Design*, arXiv Preprint, 2022.
- [5] Sizhen Li, He Zhang, Liang Zhang, **Kaibo Liu**, Boxiang Liu, et al, *LinearTurboFold: Linear-time global prediction of conserved structures for RNA homologs with applications to SARS-CoV-2 [J]*, PNAS, 2021.
- [6] **Kaibo Liu**, Boxiang Liu, He Zhang, Liang Zhang, and Liang Huang, *CoV-Seq: SARS-CoV-2 Genome Analysis and Visualization [C]*, JMIR, 2020.
- [7] Baigong Zheng, **Kaibo Liu**, Renjie Zheng, Mingbo Ma, Hairong Liu, and Liang Huang, *Simultaneous Translation Policies: From Fixed to Adaptive [C]*, ACL, 2020.
- [8] Renjie Zheng, Mingbo Ma, Baigong Zheng, **Kaibo Liu**, and Liang Huang, *Opportunistic Decoding with Timely Correction for Simultaneous Translation [C]*, ACL, 2020.
- [9] Renjie Zheng, Mingbo Ma, Baigong Zheng, **Kaibo Liu**, et al, *Fluent and Low-latency Simultaneous Speech-to-Speech Translation with Self-adaptive Training [C]*, ACL, 2020.
- [10] Mingbo Ma, Baigong Zheng, **Kaibo Liu**, Renjie Zheng, et al, *Incremental Text-to-Speech Synthesis with Prefix-to-Prefix Framework [C]*, EMNLP, 2020.
- [11] Mingbo Ma, Liang Huang, Hao Xiong, Renjie Zheng, **Kaibo Liu**, et al, *STACL: Simultaneous Translation with Implicit Anticipation and Controllable Latency using Prefix-to-Prefix Framework [C]*, In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, pp. 3025-3036. 2019.
- [12] Liang Huang, Liang, He Zhang, Dezhong Deng, Kai Zhao, **Kaibo Liu**, David Hendrix, and David Mathews, *LinearFold: linear-time approximate RNA folding by 5'-to-3' dynamic programming and beam search [C]*, Bioinformatics, 35, no. 14 (2019).
- [13] **Kaibo Liu**, Hang Lu, Zhongqing Li, et al, *Application of High Sampling Rate Data in Merging Unit for Relay Protection [C]*, 5th IEEE International Conference on Electric Utility Deregulation, Restructuring and Power Technologies, 2015, 1099-1104.
- [P1] Boxiang Liu, **Kaibo Liu**, He Zhang, etc., *Systems and methods for genome analysis and visualization [IP]*, 20220108773, 2020.
- [P2] Mingbo Ma, Liang Huang, Hao Xiong, **Kaibo Liu**, etc., *Systems and methods for simultaneous translation with integrated anticipation and controllable latency (STACL) [IP]*, 11126800, 2019.
- [P3] **Kaibo Liu**, Huanzhang Liu, Zhongqing Li, et al, *The criterion for the polarization of a single ended distance protection [IP]*, CN201510955373.9, 2015.

Awards

- Mar 2022 **General TC Technology Incentive Award 2021**, BAIDU, CHINA & USA.
- Jan 2022 **Star of Q4 2021**, BAIDU RESEARCH, USA.
- Jan 2021 **Baidu Pride Best Team Award 2020**, BAIDU, USA.
- Jul 2020 **AIG-TC Technology Incentive Award 2020-H1**, BAIDU, USA.
- Dec 2018 **AIG-TC Technology Innovation Award 2018-H2**, BAIDU, USA.
- May 2015 **First prize for scientific and technological progress in CEPRI**, STATE GRID CO. OF CHINA.
- 2010 – 2013 **National Second-order Scholarship of China**, PEKING UNIVERSITY, CHINA.

Skills & Abilities

Field	LLM (Pre-training, SFT, RL), NLP, Machine Translation, Computational biology, CV, Data analysis
LLM Stack	Pytorch, Ray, DeepSpeed, Megatron-LM/FSDP, vLLM
Programming Language	♥ Python, ♥ C/C++, MySQL, L ^A T _E X